Controlled Instrument-Relevance Sensitivity in High-Dimensional IVQR: A Monte Carlo Study

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Abstract

We study how weakening instrument relevance changes point accuracy, test calibration and inferential informativeness in high-dimensional instrumental variable quantile regression (IVQR). Extending the fixed-relevance Monte Carlo design of Chen, Huang, and Tien (2021), we scale the excluded signal and add independent compensating noise, preserving the joint treatment/outcome-disturbance distribution under the stated Gaussian construction. We compare preserved Oracle-GMM, Full-GMM and double/debiased machine learning IVQR (DML-IVQR) implementations across four relevance settings, five quantiles and sample sizes of 500 and 1000, with 500 replications per design. The contribution is a controlled sensitivity experiment, with no new estimator or theorem. At $n = 1000$, $\tau = 0.50$ and $\kappa = 0.10$, DML-IVQR has direct-at-truth acceptance 0.942, median grid-based accepted measure 3.900 out of 4, and rejection probability 0.094 at the false candidate $\alpha_0(\tau) + 0.50$. These different measures show that nominal-looking truth acceptance can coexist with little exclusion of false effects; direct acceptance is not membership coverage of the stored finite-grid set. Oracle-GMM also deteriorates under weaker relevance. DML–Full RMSE rankings have exceptions, and several strong-relevance cells exhibit substantial undercoverage. The comparisons do not isolate regularization. Non-finite mixed-density regularity, density proxies, differing residualization, same-sample fitting without cross-fitting, penalty randomness and selected-case numerical sensitivity limit interpretation; validity of the implemented nominal chi-square inference remains unestablished.

Keywords: instrumental variable quantile regression; double machine learning; weak instruments; Monte Carlo simulation; confidence-set informativeness

1 Introduction

Endogenous treatment makes identification of structural quantile effects distinct from ordinary conditional quantile regression (Koenker and Bassett 1978; Chernozhukov and Hansen 2005). IVQR uses excluded instruments and structural-rank restrictions, but valid moments alone need not distinguish competing treatment coefficients (Chernozhukov, Hansen, and Wüthrich 2017). Candidate-specific inversion accommodates weak identification under the conditions of the theoretical procedures (Chernozhukov and Hansen 2008); weak relevance can nevertheless leave many false candidates unexcluded. Conventional first-stage diagnostics provide no universal IVQR identification threshold.

Many controls introduce a separate nuisance-estimation problem. Penalized quantile regression and orthogonal-score methods reduce sensitivity to nuisance errors under their respective conditions (Belloni and Chernozhukov 2011; Belloni, Chernozhukov, and Kato 2019; Chernozhukov, Chetverikov, et al.

2018), but do not create identifying variation. Chen et al. (2021) combine regularized nuisance estimation and an orthogonal IVQR score in a high-dimensional Monte Carlo design. Their theoretical construction motivates this study; our empirical comparisons concern preserved executable implementations.

The high-dimensional simulation of Chen et al. (2021) holds instrument relevance fixed. It does not trace performance as relevance weakens while preserving the treatment distribution and its relationship with the outcome disturbance. We address this gap by scaling the excluded signal and adding independent compensating noise. Under the stated Gaussian construction, the modification preserves the joint distribution of treatment and the outcome disturbance while changing treatment–instrument relationships. This controls treatment dispersion, without isolating computational mechanisms behind estimator differences.

The scientific question extends beyond point accuracy. Direct-at-truth acceptance can approach the nominal confidence level while most of the grid is retained and few false candidates are rejected; a more accurate point estimate also need not imply better calibration. Reported coverage is a direct-test acceptance diagnostic, not literal membership coverage of stored grid sets. The distinct evaluation rules are specified in Section 3.2.

The paper makes three empirical contributions. First, it specifies a controlled Monte Carlo relevance modification preserving the stated treatment/endogeneity distribution. Second, it compares preserved Oracle-GMM, Full-GMM and DML-IVQR implementations across relevance settings and quantiles: estimators share the observed sample within each setting, while relevance settings reuse primitive draws but generally have different realized treatment and outcome values. This reuse does not make the DML pivotal penalty draws common across evaluations. Third, it jointly evaluates point accuracy, direct-at-truth acceptance, false-candidate rejection and finite-grid informativeness. No new estimator or inference theorem is proposed.

The implementation-level RMSE comparisons have exceptions. Differences in nuisance dimension, residualization and computational details prevent attributing this comparison to regularization alone. Strong-relevance undercoverage limits any reading of nominal-looking acceptance as validated inference.

The findings concern one DGP family and historical implementations with a non-finite mixed residual density, density proxies, differing residualization, same-sample fits and fresh penalty draws. Implemented orthogonality, nominal chi-square validity and transfer of theoretical weak-identification robustness remain unestablished; missing cross-fitting alone does not prove invalidity. Appendix D documents selected-case rank/decision sensitivity, not a full-experiment error frequency. Historical results are retained, and finite grids establish no unrestricted confidence-set geometry.

2 Econometric Framework

2.1 Structural moments and orthogonalization

Let Y_i be the outcome, D_i endogenous treatment, X_i controls, and $Z_i \in \mathbb{R}^{d_z}$ instruments. Under the structural-rank assumptions of Chernozhukov and Hansen (2005), the monotone structural quantile function has the representation

$$Y_i = q(U_i, D_i, X_i), \qquad U_i \mid X_i, Z_i \sim \text{Uniform}(0,1), \tag{1}$$

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau). \tag{2}$$

The treatment coefficient $\alpha_0(\tau)$ is the target. The intercept and control coefficients are nuisances. Rank similarity permits treatment-state-specific ranks with the same relevant conditional distribution; rank invariance is the stronger common-rank case. The structural quantile at fixed treatment generally differs from the observational quantile conditional on endogenous treatment. Monotonicity and conditional rank uniformity imply

$$\mathrm{E}\big[\tau - \mathbf{1}\{Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau)\} \;\big|\; X_i, Z_i\big] = 0. \tag{3}$$

Instrument validity supports this restriction; identification additionally requires variation that distinguishes alternative treatment coefficients.

For candidate a, define $\widetilde{X}_i = (1, X_i')'$, $\rho_\tau(v) = v[\tau - \mathbf{1}\{v < 0\}]$, and a population nuisance profile (under the reference conditions, assume uniqueness or a fixed measurable selection from the minimizers)

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg\min_{c,b} \mathrm{E}\left[\rho_\tau\left(Y_i - D_i'a - c - X_i'b\right)\right]. \tag{4}$$

$$\epsilon_i(a) = Y_i - D_i'a - c(a) - X_i'\beta(a). \tag{5}$$

$$\begin{aligned} M(a) &= \mathrm{E}\left[Z_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\right], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathrm{E}\left[\widetilde{X}_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\right], \quad J(a) \in \mathbb{R}^{(p+1)\times(p+1)}. \end{aligned} \tag{6}$$

$$\delta(a) = M(a)J(a)^{-1}, \qquad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \tag{7}$$

$$\Psi_i(a) = Z_i - \delta(a)\widetilde{X}_i = Z_i - d_0(a) - \Gamma(a)X_i. \tag{8}$$

Here $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ is the conditional residual density at zero. With $\eta = (c, \beta', \mathrm{vec}(\delta)')'$, let $\eta(a)$ collect the candidate-specific profile and $\eta_0 = \eta(\alpha_0(\tau))$. Let $W_i = (Y_i, D_i', X_i', Z_i')'$ collect an observation; it is distinct from the scalar candidate statistic $W_{e,n}(a)$ below. The ideal score is a theoretical reference:

$$\begin{aligned} g_\tau(W_i; a, \eta) &= \left[\tau - \mathbf{1}\left\{Y_i \le D_i'a + c + X_i'\beta\right\}\right]\left(Z_i - \delta\widetilde{X}_i\right), \\ g_\tau(W_i; a, \eta(a)) &= \left[\tau - \mathbf{1}\left\{Y_i \le D_i'a + c(a) + X_i'\beta(a)\right\}\right]\Psi_i(a), \end{aligned} \tag{9}$$

$$\mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \tag{10}$$

$$\partial_\eta \,\mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} = 0. \tag{11}$$

Under finite density-weighted moments, a nonsingular J, and the requisite differentiation conditions, the derivative in the intercept and slopes is $-[M_0 - \delta_0 J_0] = 0$. The residualization derivative vanishes by the quantile moment. Appendix D.7 gives the algebra. These are properties of the ideal construction under its conditions, rather than guarantees for every executable implementation.

This distinction is material for the present DGP. At truth, the outcome residual is $e_\tau = D(u - q_\tau)$, with $q_\tau = \Phi^{-1}(\tau)$ and $D = \Phi(T)$, $T = d^*(\kappa)$. Its density conditional on (X, Z) mixes over treatment and equals $\phi(q_\tau)\,\mathrm{E}[1/\Phi(T) \mid u = q_\tau, X, Z]$. The conditional normal variance $2.91 - \kappa^2 > 1$ makes this expectation infinite at every investigated setting. Appendix D.6 gives the retained Gaussian conditioning and negative-tail argument; this is not the density conditional on realized treatment. The experiment therefore does not establish the ideal density-weight regularity conditions or validate the associated asymptotic theory.

2.2 Preserved estimators and inference

Oracle-GMM uses $X_1, \ldots, X_{10}$, the designated oracle set. Full-GMM uses all 100 controls. Both fit an unpenalized candidate-specific quantile regression with an intercept, then directly invert density-proxy-weighted control moment matrices without a residualization intercept. DML-IVQR fits quantile Lasso and regularized instrument regressions using all controls. Its Belloni–Chernozhukov pivotal penalty follows the executable route of Chen et al. (2021); their comparison with cross-validation motivates retaining this route.

The quantile-Lasso objective is an unscaled sum of check losses plus one half of the supplied coefficient-specific penalty vector times absolute coefficients, including the intercept. There is no post-Lasso refit. Instrument regressions use square-root-density-transformed instruments and controls with an ordinary unpenalized intercept. The score subtracts the retained intercept and original-control fitted slopes from the untransformed instrument. This intercept differs from the constant regressor in the ideal weighted

projection. The fitted-normal proxy is evaluated at candidate residuals; the preserved R call `dnorm(e, mean(e), var(e))` supplies the residual variance where R interprets a standard deviation. It is not a direct estimate of the theoretical conditional density at zero.

Nuisances and scores use the same observations, with no sample splitting or cross-fitting. Fresh pivotal draws are generated at each penalty call, introducing algorithmic randomness across candidate evaluations. Appendix D records the penalty formulas, residualization rules, and environment details. The estimator comparison encompasses these implementation differences as well as regularization and nuisance dimension; it cannot isolate the effect of regularization alone.

These differences prevent automatic transfer of ideal orthogonality or inference theory; same-sample justification remains unestablished (Appendix D.7). Absence of cross-fitting alone does not prove invalidity.

For estimator e, let $\widehat{g}_{e,i}(a)$ be its score. The common criterion uses the uncentered score second moment:

$$\widehat{g}_{e,n}(a) = \frac{1}{n}\sum_{i=1}^{n} \widehat{g}_{e,i}(a). \tag{12}$$

$$\widehat{\Sigma}_e(a) = \frac{1}{n}\sum_{i=1}^{n} \widehat{g}_{e,i}(a)\widehat{g}_{e,i}(a)'. \tag{13}$$

$$W_{e,n}(a) = n\widehat{g}_{e,n}(a)'\widehat{\Sigma}_e(a)^{-1}\widehat{g}_{e,n}(a). \tag{14}$$

$$\widehat{\alpha}_e(\tau) = \arg\min_{a\in\mathcal{A}} W_{e,n}(a). \tag{15}$$

The Monte Carlo candidate set $\mathcal{A}$ is the fixed grid specified below; the first candidate attaining the minimum stored grid statistic is retained, and an incomplete profile receives no point estimate. Direct inversions use neither a generalized inverse nor additional matrix regularization. Failures or non-finite statistics are recorded as numerical failures.

The procedures use the asymptotic $\chi^2_{d_z}$ reference cutoff nominally. For the stored candidate evaluations on the finite grid $\mathcal{A} = G_0$ specified below, inversion at nominal significance level γ gives the grid accepted set

$$C^{\text{grid}}_{e,1-\gamma}(\tau) = \left\{a \in \mathcal{A} : W_{e,n}(a) \le c_{1-\gamma,d_z}\right\}. \tag{16}$$

This set is defined by the stored grid evaluations, not by separately rerun direct tests. Conceptually, unrestricted test inversion would retain all real candidate values accepted by a fully specified candidate-test rule, including its randomization. That unrestricted set is not computed here. Direct-at-truth test acceptance, defined in Section 3.2, is a third object; its reported frequency is not literal membership coverage of the stored finite-grid set.

Here $c_{1-\gamma,d_z}$ is the chi-square quantile, with $d_z = 2$. Its validity for the preserved implementations under the studied DGP has not been established. The cited theoretical candidate tests can accommodate weak identification under their regularity conditions (Chernozhukov and Hansen 2008), but that robustness does not automatically transfer to the implemented density proxies, residualization and nuisance fits. This construction differs from exact conditional finite-sample inference (Chernozhukov, Hansen, and Jansson 2009). Weak-identification robustness does not ensure narrow sets or high power, and a finite numerical domain cannot establish global boundedness.

3 Monte Carlo Design

3.1 Baseline and controlled relevance extension

The baseline follows the executable high-dimensional design of Chen et al. (2021), with independent observations and 100 controls. The outcome and assignment disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \tag{17}$$

$$x_{ji} \overset{\text{ind}}{\sim} \mathcal{N}(0,1), \quad j = 1, \ldots, 100, \qquad z_{1i}, z_{2i}, v_{1i}, v_{2i} \overset{\text{ind}}{\sim} \mathcal{N}(0,1). \tag{18}$$

$$X_{ji} = \Phi(x_{ji}), \qquad j = 1, \ldots, 100, \tag{19}$$

The remaining primitives are mutually independent and independent of (u_i, ϵ_i). The controls $X_{ji} = \Phi(x_{ji})$ are independent Uniform$(0,1)$ variables. The observed instruments are

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \tag{20}$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \tag{21}$$

The released code uses transformed X controls in these equations, although the source paper displays corresponding terms as lowercase x. We retain the executable construction. The baseline treatment and outcome are

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \tag{22}$$

$$Y_i = 1 + D_i + 5 \sum_{j=1}^{7} X_{ji} + D_i u_i. \tag{23}$$

The correlation of u_i with ϵ_i generates endogeneity through the treatment index and the interaction $D_i u_i$. Seven controls enter the outcome directly, and the union of outcome and instrument controls is the ten-control oracle set. At fixed $d \in (0,1)$,

$$Y_i(d) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d(1 + u_i). \tag{24}$$

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d\left[1 + \Phi^{-1}(\tau)\right]. \tag{25}$$

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \tag{26}$$

The five structural effects are approximately -0.282, 0.326, 1.000, 1.674, and 2.282 in increasing quantile order.

The original design fixes relevance. To reduce the excluded signal without simultaneously changing treatment dispersion, introduce $w_i \sim \mathcal{N}(0,1)$ independently of all baseline variables and set

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1-\kappa^2)}\, w_i + \epsilon_i, \tag{27}$$

$$D_i(\kappa) = \Phi\big(d_i^*(\kappa)\big). \tag{28}$$

$$\mathrm{Var}\big[d_i^*(\kappa)\big] = 2\kappa^2 + 2(1-\kappa^2) + 1 = 3. \tag{29}$$

$$\mathrm{Cov}\big(Z_{1i}, d_i^*(\kappa)\big) = \mathrm{Cov}\big(Z_{2i}, d_i^*(\kappa)\big) = \kappa. \tag{30}$$

$$\mathrm{Cov}\big(u_i, d_i^*(\kappa)\big) = \mathrm{Cov}(u_i, \epsilon_i) = 0.3. \tag{31}$$

Joint normality implies $d_i^*(\kappa) \sim \mathcal{N}(0,3)$ and an invariant joint distribution of $(u_i, d_i^*(\kappa))$, hence of $(u_i, D_i(\kappa))$. The construction preserves the marginal treatment distribution, the relationship generating endogeneity, the outcome equation, structural effects, controls, and instrument construction. It deliberately

changes treatment–instrument relationships. Appendix B derives the compensating coefficient and these invariances. At $\kappa = 1$ the baseline is recovered; the conceptual endpoint $\kappa = 0$ removes the primitive excluded signal. The smallest simulated value is positive, so it is not a formal non-identification experiment. The parameter orders designed relevance and is neither an F-statistic nor a universal measure of quantile-specific identification.

3.2 Design dimensions and performance measures

$$\begin{aligned} &n \in \{500, 1000\}, && \tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ &\kappa \in \{1, 0.50, 0.25, 0.10\}, && R = 500. \end{aligned} \tag{32}$$

The 40 designs yield 120 estimator–design cells. Within each replication, primitive draws are reused across relevance settings and estimators use the same observed sample. This common-random-number design improves comparison but does not eliminate Monte Carlo uncertainty. Inference uses a 95% level. A separate 100-replication calibration reports nested OLS F-statistics and partial R^2 for adding both instruments to a regression of treatment on an intercept and the ten oracle controls; these are descriptive diagnostics without an IVQR threshold.

For replication r, point accuracy is summarized by

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R}\sum_{r=1}^{R} \left[\alpha_0(\tau) - \widehat{\alpha}_r(\tau)\right], \\ \text{MAE}(\tau) &= \frac{1}{R}\sum_{r=1}^{R} \left|\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right|, \\ \text{RMSE}(\tau) &= \left\{\frac{1}{R}\sum_{r=1}^{R} \left[\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right]^2\right\}^{1/2}. \end{aligned} \tag{33}$$

MAE and RMSE denote mean absolute error and root mean squared error. The preserved bias convention is truth minus estimate: positive bias indicates underestimation. The quantity historically labeled coverage is direct-at-truth candidate-test acceptance, used here as a test-inversion coverage diagnostic. Power is evaluated at specified false candidates. Both use direct evaluations, independently of grid membership; the computed quantities retain their original definitions:

$$C_r(\tau) = \mathbf{1}\left\{W_{n,r}(\alpha_0(\tau)) \le c_{0.95}\right\}, \qquad \widehat{\text{Coverage}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} C_r(\tau). \tag{34}$$

$$\widehat{\text{Size}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\left\{W_{n,r}(\alpha_0(\tau)) > c_{0.95}\right\} = 1 - \widehat{\text{Coverage}}(\tau). \tag{35}$$

$$a_\Delta(\tau) = \alpha_0(\tau) + \Delta, \qquad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \tag{36}$$

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\left\{W_{n,r}(a_\Delta(\tau)) > c_{0.95}\right\}. \tag{37}$$

For DML, fresh pivotal draws are generated for each penalty call, so a direct-truth evaluation and a stored-profile evaluation can use different draws even at the same candidate. Reported coverage is the repeated-sample acceptance frequency of the direct test, not literal membership coverage of the stored finite-grid set $C^{\text{grid}}_{e,0.95}(\tau)$. Except at the median, the unrounded structural truths do not belong to G_0; even at an on-grid truth, the separate DML computations need not give the same acceptance decision. Conceptual unrestricted inversion requires a fully specified candidate-test rule, including its randomization, and is not reconstructed from these separately recorded evaluations. Raw power uses the common nominal cutoff without size adjustment.

The cutoff and common primary grid are

$$c_{0.95} = \chi^2_{2,0.95} = 5.991464547\ldots. \tag{38}$$

$$\begin{aligned} A_0 &= [-1, 3], && h = 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \ldots, 41\}, \\ &= \{-1, -0.9, \ldots, 2.9, 3\}. \end{aligned} \tag{39}$$

The four-unit domain and 0.1 resolution retain the original executable grid for all designs. All five true effects lie strictly inside it. Grid point estimates include discretization, endpoint, and first-tie effects; the grid is a computational domain, not an asserted theoretical parameter space.

Let $I_r(a_j) = \mathbf{1}\{W_{n,r}(a_j) \le c_{0.95}\}$. The accepted-set measure is

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} \left[I_r(a_j) + I_r(a_{j+1}) \right]. \tag{40}$$

This trapezoidal sum of binary acceptance indicators ranges from zero to four. Adjacent intervals contribute 0, 0.05, or 0.10; the calculation summarizes grid acceptance, allowing multiple accepted grid runs, without locating continuous boundaries. It is a quadrature summary, not the Lebesgue length of the finite accepted set, the span between extreme accepted values, or unrestricted confidence-region length. Full-grid acceptance requires acceptance at all 41 points; either-boundary acceptance requires acceptance at at least one endpoint. Neither event establishes acceptance between all points or outside the domain.

Appendix A examines $A_1 = [-3, 5]$ and $A_2 = [-5, 7]$ at spacing 0.1 in a separate 100-replication diagnostic with $n = 1000$, all quantiles and estimators, and $\kappa \in \{1, 0.25, 0.10\}$. It reuses saved samples and interior profiles while adding candidates. Coverage and power remain separate direct evaluations. This bounded diagnostic assesses numerical localization over investigated ranges only.

4 Simulation Results

Appendix D.2 maps the principal numerical findings, including comparisons using all design cells, to the exact frozen CSVs and existing full-table and figure assets. These sources permit inspection without re-estimation.

4.1 Instrument-Relevance Calibration

Table 1 checks whether the four values of κ produce the intended ordering of excluded-instrument relevance. The nested OLS diagnostic specified in Section 3.2 uses 100 replications for each (n, κ) design. They are descriptive first-stage diagnostics rather than formal IVQR identification statistics.

Both diagnostics decrease at every step as κ falls. For $n = 500$, the median F-statistic declines from 101.17 at $\kappa = 1$ to 18.71, 4.74, and 1.22 at the three weaker settings. For $n = 1000$, the corresponding sequence is 212.82, 38.65, 9.85, and 2.04. The dispersion ranges in Table 1 shift in the same direction. Median partial R^2 falls from 0.294 to 0.005 for $n = 500$ and from 0.301 to 0.004 for $n = 1000$, with intermediate values near 0.07 at $\kappa = 0.50$ and 0.02 at $\kappa = 0.25$.

At a fixed κ, partial R^2 is similar across the two sample sizes because it records the incremental explanatory share of the two instruments. The F-statistic nevertheless rises with n because that contribution is estimated more precisely with more observations. The joint movement of the two diagnostics therefore confirms the designed relevance ordering: $\kappa = 1$ provides the strongest relationship and $\kappa = 0.10$ the weakest. This ordering is used to organise the results below; it does not impose a conventional first-stage threshold or establish a universal IVQR identification classification.

Table 1. Instrument-strength calibration

n	κ	Median F	F: p10–p90	Median partial R^2	Partial R^2: p10–p90
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo instrument-relevance diagnostics. The reported dispersion ranges are the 10th and 90th percentiles.

4.2 Point-Estimation Accuracy

Table 2 displays finite-sample performance for $n = 1000$, the representative quantiles $\tau \in \{0.10, 0.50, 0.90\}$, all four κ values, and the three estimators. The comparisons in this subsection also use the complete saved results for both sample sizes and all five quantiles. Each point estimate is the first minimiser of the candidate-specific statistic on the ordered 41-point grid in $A_0 = [-1, 3]$, rather than a continuously optimised estimate.

Point accuracy deteriorates systematically as relevance weakens. Across the complete results, both MAE and RMSE increase strictly at every step from $\kappa = 1$ to 0.10 in all 30 estimator–sample-size–quantile sequences. For example, at $n = 1000$ and $\tau = 0.50$, Oracle RMSE rises from 0.072 through 0.134 and 0.431 to 0.945. The corresponding strong-to-weak changes are 0.100 to 1.018 for Full-GMM and 0.104 to 0.991 for DML-IVQR. Deterioration of Oracle-GMM, which knows the designated relevant control set, shows that high-dimensional nuisance estimation is not the sole source of the loss of accuracy. It does not identify which other feature of the design or implementation produces the magnitude of that loss.

Increasing the sample size improves RMSE in every one of the 60 matched estimator–quantile–κ comparisons. At $\tau = 0.50$, Oracle RMSE falls from 0.102 to 0.072 when n rises from 500 to 1000 at $\kappa = 1$, and from 1.027 to 0.945 at $\kappa = 0.10$. At $\tau = 0.90$ and $\kappa = 0.50$, DML RMSE falls from 0.539 to 0.396. These reductions show lower observed Monte Carlo RMSE at the larger sample size, but the remaining errors can still be substantial under weak relevance. Two sample sizes do not establish an asymptotic rate or imply that inference is well calibrated or informative in every design.

Estimator rankings are systematic but not uniform. Across the 40 matched (n, τ, κ) designs, the preserved DML-IVQR implementation has lower RMSE than the preserved Full-GMM implementation in 35 and lower RMSE than the preserved Oracle-GMM implementation in 11. A sizeable DML advantage occurs at $n = 500$, $\tau = 0.90$, and $\kappa = 0.50$, where RMSE is 0.539 for DML, 1.362 for Full, and 0.822 for Oracle. One clear exception is $n = 500$, $\tau = 0.10$, and $\kappa = 0.50$, where DML RMSE is 1.335 compared with 1.219 for Full. Other reversals are small: at $n = 1000$, $\tau = 0.50$, and $\kappa = 1$, the values are 0.104 and 0.100. These counts are descriptive, and such small differences do not establish statistically meaningful superiority. Oracle is a benchmark with knowledge of the relevant controls, not a guaranteed finite-sample winner; moreover, the estimators differ in residualisation and other implementation details as well as regularisation.

Quantile patterns also distinguish directional bias from overall error. Under the convention Bias = $\alpha_0 - \widehat{\alpha}$, negative bias means overestimation and positive bias means underestimation. Across both sample sizes, all estimators, and all four κ values, the 24 biases at $\tau = 0.10$ are negative, whereas the 24 biases at $\tau = 0.90$ are positive. The estimates therefore move inward on average at the two tail quantiles. No equally uniform sign statement holds for the three interior quantiles. Using unrounded values, the median has lower MAE and lower RMSE than each tail in every fixed-estimator, fixed-n, fixed-κ comparison. This uniform centre-versus-tail ordering in point accuracy does not extend to coverage, which varies differently across quantiles and designs. Mean bias can also conceal large replication errors: for Oracle at $n = 1000$,

Table 2. Main finite-sample performance at $n = 1000$

τ	κ	Estimator	Bias	MAE	RMSE	Coverage	Median accepted-set measure
0.10	1.00	Oracle-GMM	-0.054	0.093	0.121	0.860	0.400
		Full-GMM	-0.329	0.337	0.387	0.502	0.800
		DML-IVQR	-0.005	0.117	0.150	0.968	0.600
	0.50	Oracle-GMM	-0.101	0.194	0.287	0.926	1.000
		Full-GMM	-0.407	0.465	0.630	0.804	1.900
		DML-IVQR	-0.159	0.306	0.482	0.948	2.225
	0.25	Oracle-GMM	-0.384	0.566	0.934	0.930	3.450
		Full-GMM	-0.773	0.914	1.242	0.932	3.500
		DML-IVQR	-0.681	0.844	1.229	0.950	3.750
	0.10	Oracle-GMM	-0.747	0.975	1.376	0.946	4.000
		Full-GMM	-1.074	1.251	1.582	0.942	3.900
		DML-IVQR	-0.932	1.146	1.514	0.936	4.000
0.50	1.00	Oracle-GMM	0.007	0.047	0.072	0.872	0.200
		Full-GMM	0.007	0.072	0.100	0.896	0.400
		DML-IVQR	0.070	0.079	0.104	0.776	0.300
	0.50	Oracle-GMM	0.015	0.093	0.134	0.936	0.500
		Full-GMM	0.014	0.151	0.199	0.946	0.800
		DML-IVQR	0.059	0.119	0.160	0.918	0.600
	0.25	Oracle-GMM	0.045	0.281	0.431	0.946	1.900
		Full-GMM	0.029	0.372	0.500	0.936	2.200
		DML-IVQR	-0.003	0.306	0.448	0.948	2.025
	0.10	Oracle-GMM	-0.044	0.735	0.945	0.942	3.900
		Full-GMM	-0.012	0.840	1.018	0.948	3.800
		DML-IVQR	-0.195	0.779	0.991	0.942	3.900
0.90	1.00	Oracle-GMM	0.072	0.110	0.148	0.904	0.500
		Full-GMM	0.387	0.397	0.465	0.530	1.000
		DML-IVQR	0.183	0.186	0.215	0.498	0.400
	0.50	Oracle-GMM	0.211	0.295	0.540	0.940	1.575
		Full-GMM	0.584	0.629	0.871	0.834	2.450
		DML-IVQR	0.226	0.260	0.396	0.844	1.100
	0.25	Oracle-GMM	0.491	0.658	1.038	0.952	3.650
		Full-GMM	0.901	1.058	1.389	0.920	3.600
		DML-IVQR	0.417	0.580	0.927	0.922	3.350
	0.10	Oracle-GMM	0.653	0.948	1.350	0.964	4.000
		Full-GMM	1.050	1.236	1.583	0.948	3.900
		DML-IVQR	0.519	0.831	1.191	0.940	4.000

Note: Each cell is based on 500 Monte Carlo replications. Bias follows the frozen convention (Bias = $\alpha_0 - \widehat{\alpha}$). Coverage denotes direct-at-truth test acceptance, not membership coverage of the stored grid set. The accepted-set measure is the grid-based measure within the prespecified primary computational domain ($A_0 = [-1, 3]$); it is not an unrestricted confidence-region length.

$\tau = 0.50$, and $\kappa = 0.10$, Bias is only -0.044 while RMSE is 0.945. Positive and negative errors partly cancel in the mean, so small Bias is not evidence of small MAE or RMSE.

The fixed grid affects the point estimator through its resolution, endpoints, and first-tie rule. Yet no diagnostic isolates those constraints or regularisation as the cause of the bias pattern.

4.3 Direct-at-Truth Acceptance and Finite-Sample Calibration

Figure 1 shows that direct-at-truth acceptance (historically labelled coverage) differs considerably from the nominal 95% level, even under the strongest instrument relevance considered. At $n = 1000$ and $\kappa = 1$, Oracle coverage is 0.872 at the median, while Full-GMM coverage is only 0.502 and 0.530 at $\tau = 0.10$ and 0.90. DML displays marked tail asymmetry, with coverage 0.968 at $\tau = 0.10$ and 0.498 at $\tau = 0.90$. Figure 1 reports the complete estimator–quantile pattern. Because coverage is evaluated directly at $\alpha_0(\tau)$, these discrepancies are not generated by rounding truth to the numerical grid.

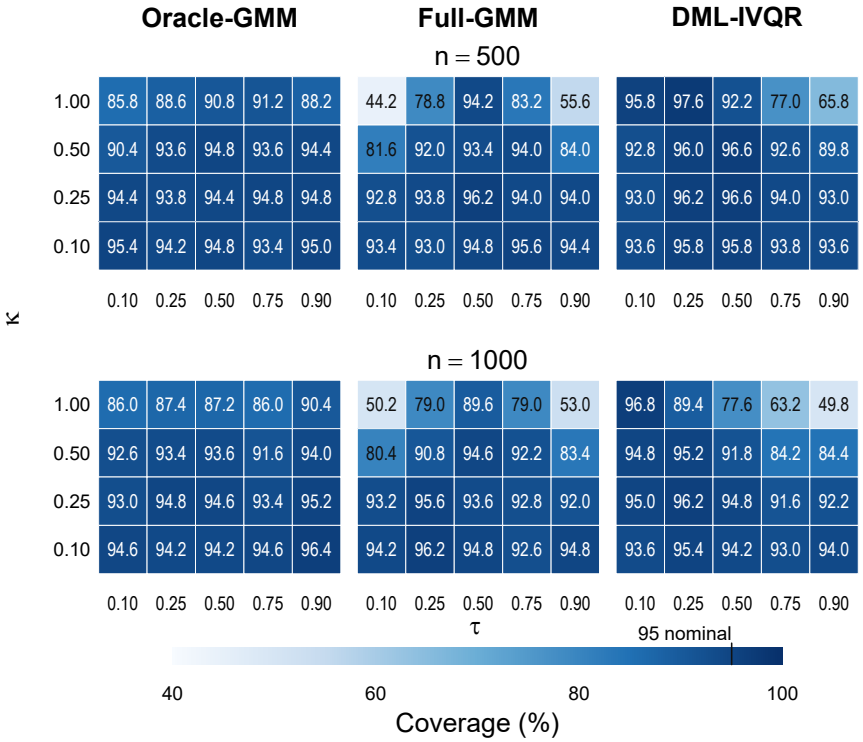


Figure 1. Empirical coverage across simulation designs.

Note: Nominal coverage is 95%; each cell uses 500 complete replications. Coverage is the fraction of tests accepting the true parameter, evaluated directly at that parameter rather than reconstructed from grid membership. Panel columns identify estimators; panel rows show $n = 500$ and $n = 1000$. Within each panel, columns show quantiles τ and rows show relevance κ, declining from 1.00 to 0.10. Annotations and the common colour scale report percentages. Darker shading indicates higher coverage, not necessarily better calibration.

Increasing the sample size can improve point accuracy without improving calibration. At $\kappa = 1$, DML coverage falls from 0.976 to 0.894 between $n = 500$ and $n = 1000$ at $\tau = 0.25$, from 0.922 to 0.776 at $\tau = 0.50$, from 0.770 to 0.632 at $\tau = 0.75$, and from 0.658 to 0.498 at $\tau = 0.90$. These changes equal -8.2, -14.6, -13.8, and -16.0 percentage points. Section 4.2 found lower RMSE at $n = 1000$ in all 60 matched comparisons, so lower observed RMSE of the grid minimiser does not imply that the candidate-specific test has size closer to its nominal level.

Coverage tends to move toward 0.95 as relevance weakens, but not monotonically in every cell. Averaged over estimators, quantiles, and sample sizes, coverage rises from 0.798 at $\kappa = 1$ to 0.912, 0.941, and 0.944 as κ falls. Only 15 of the 30 estimator–sample-size–quantile sequences are nondecreasing at every step. For example, DML coverage at $n = 1000$ and $\tau = 0.10$ is 0.968, 0.948, 0.950, and 0.936 across declining κ, while Full-GMM coverage at $n = 500$ and $\tau = 0.25$ is 0.788, 0.920, 0.938, and 0.930.

These reversals are generally small compared with the strong-relevance distortions. More importantly, movement toward nominal coverage under weak relevance can result from rejecting candidate values less often overall.

Each rate is based on $R = 500$ replications. Under independent replications, its conventional plug-in Monte Carlo standard error is

$$\mathrm{MCSE}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{R}}.$$

The maximum MCSE is approximately 0.022 and the MCSE at 0.95 is approximately 0.010. Appendix C strengthens this cell-by-cell uncertainty check. Across 60,000 valid direct-at-truth evaluations, 53 pointwise two-sided 95% Wilson intervals lie wholly below 0.95, one lies wholly above it, and 66 contain it. No truth statistic is missing or nonfinite. The intervals use integer acceptance counts without a continuity correction; they are pointwise Monte Carlo intervals, not simultaneous or multiplicity-adjusted tests. An individual rate's MCSE is not used as the standard error of a difference between correlated simulation results. These uncertainty summaries assume independent replications; historical DML worker-stream independence is not verified (Appendix D.3).

The audit rules out missing-value handling, nonfinite truth statistics, and reconstruction from rounded grid confidence sets as explanations for the observed undercoverage. It does not establish invariance to residual-rank arithmetic or numerical solution details; later selected-case audits document such sensitivity, as summarized in Appendix D. It does not isolate a cause among finite-sample approximation error, features of the preserved implementation, or failure of the idealised regularity conditions for the asymptotic chi-square reference law in this DGP. Strong-relevance undercoverage also shows that weak relevance alone cannot explain the distortions. Coverage records retention of the truth; it does not record how readily false candidates are excluded. The next two sections therefore examine power and accepted-set informativeness before interpreting near-nominal coverage under weak relevance.

4.4 Power Against False Candidate Values

Figure 2 reports rejection probabilities at $\tau = 0.50$ for candidate values $a_\Delta(\tau) = \alpha_0(\tau) + \Delta$. At $\Delta = 0$, the candidate is true and rejection probability is empirical size; at each of the six nonzero deviations, it is empirical power against that specified false candidate. The data-generating parameter remains fixed throughout. The plotted probabilities use the same nominal 5% chi-square cutoff and are not adjusted for differences in empirical size.

At $n = 1000$ and $\Delta = -0.50$, Oracle power falls from 1.000 through 0.850 and 0.316 to 0.082 as κ declines, while DML falls from 1.000 through 0.802 and 0.312 to 0.104. Full-GMM likewise falls from 0.998 to 0.076 between the strongest and weakest settings. For $\Delta = 0.50$ at $\kappa = 0.25$, power is 0.378, 0.238, and 0.286 for Oracle, Full, and DML, before falling to approximately one tenth for all three at $\kappa = 0.10$. The deterioration is therefore already substantial at intermediate relevance, and false effects separated from truth by one half become difficult to exclude in the weakest setting.

Increasing n usually raises power where usable signal remains. At $\kappa = 0.50$, Oracle power at $\Delta = -0.50$ rises from 0.552 to 0.850, Full from 0.260 to 0.634, and DML from 0.466 to 0.802 between $n = 500$ and $n = 1000$. At $\Delta = 0.50$, the corresponding changes are 0.560 to 0.940, 0.276 to 0.726, and 0.392 to 0.852. At $\kappa = 0.10$, improvements are much smaller and not uniform: Full power at $\Delta = -0.50$ falls from 0.088 to 0.076, while DML remains 0.104; at $\Delta = 0.50$, DML rises from 0.030 to 0.094. Across all five quantiles and six nonzero deviations, $n = 1000$ has higher rejection probability in 305 of 360 matched cells, the same probability in 25, and a lower probability in 30.

The null points clarify why raw power cannot be used as an efficiency ranking. At $n = 1000$ and $\kappa = 1$, median empirical size is 0.128 for Oracle, 0.104 for Full, and 0.224 for DML, compared with 0.092, 0.058, and 0.078 at $n = 500$. DML's strong-relevance rejection of false candidates therefore coexists with the size distortion documented in Section 4.3. At $n = 1000$ and $\kappa = 0.10$, empirical sizes are 0.058, 0.052, and 0.058, while power at $\Delta = -0.50$ is only 0.082, 0.076, and 0.104. Rejection of this false candidate is then close to rejection at truth, indicating little discrimination despite approximately nominal size.

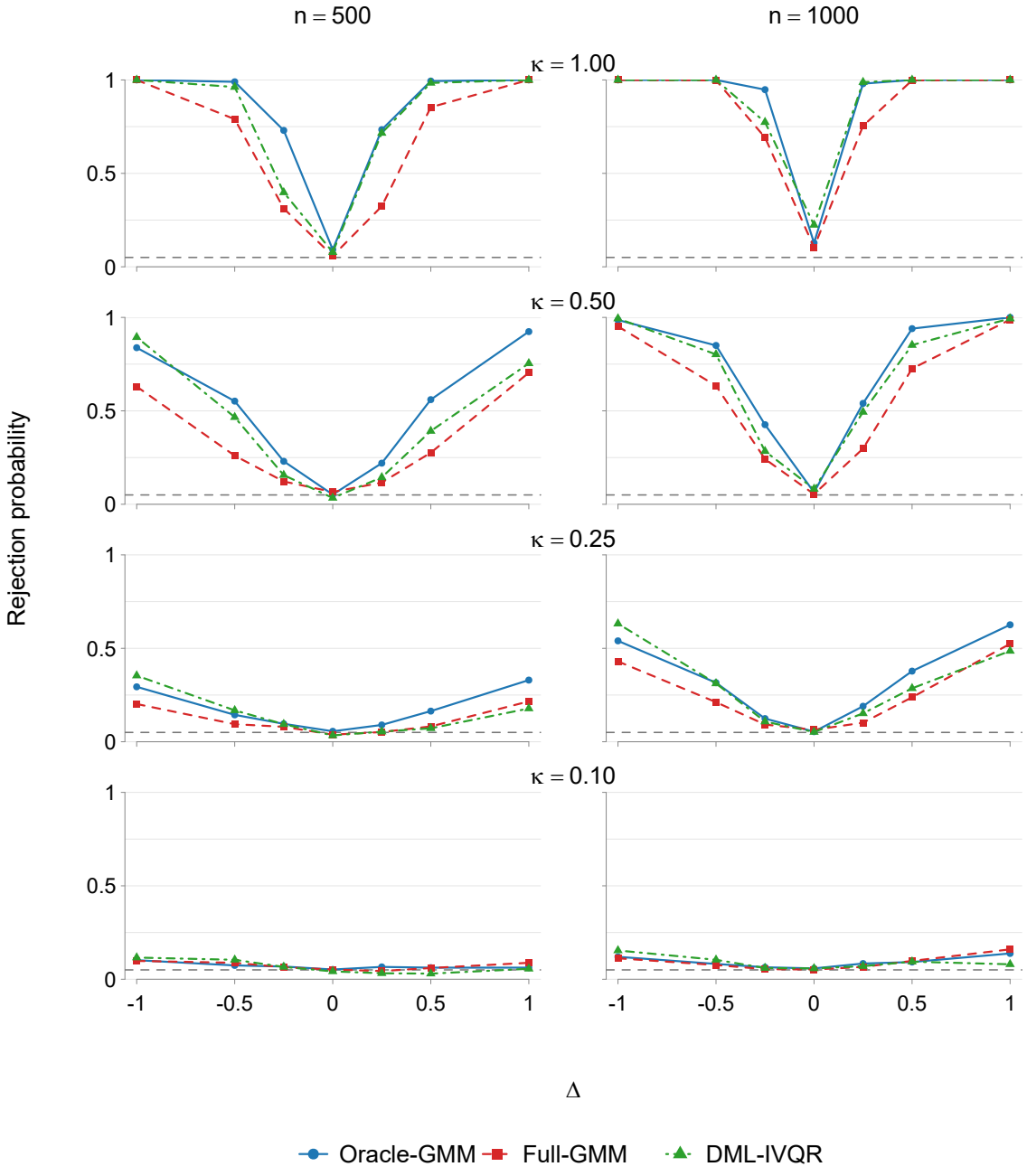


Figure 2. Empirical power at the median.

Note: The median benchmark is $\tau = 0.50$. Columns show $n = 500$ and $n = 1000$; rows show $\kappa = 1.00, 0.50, 0.25, 0.10$. The horizontal axis is $\Delta = a - \alpha_0(0.50)$, the saved deviation of the tested candidate from truth. Each point uses 500 complete replications and the same nominal 5% chi-square cutoff; dashed horizontal lines mark 0.05. Rejection at truth is evaluated directly. At $\Delta = 0$, rejection probability measures empirical size; at nonzero deviations, it measures power. These rejection probabilities are not size-adjusted. Straight segments are visual guides between evaluated points.

The median pattern broadly generalises, but it is not universally monotone. Across all quantiles, sample sizes, estimators, and nonzero deviations, power is nonincreasing as κ falls in 162 of 180 sequences. The 18 exceptions are mostly small reversals when rejection is already close to empirical size. Signed differences also occur: for DML at $\tau = 0.50$, $n = 500$, and $\kappa = 0.25$, power is 0.168 at $\Delta = -0.50$ but 0.072 at $\Delta = 0.50$. Figure 2 displays only $\tau = 0.50$, so these full-result counts support generalisation without implying that every quantile follows the plotted paths. The design does not isolate the source of signed asymmetry, and asymmetry in power is not by itself evidence of point-estimation bias.

4.5 Confidence-Set Informativeness

Figure 3 reports the median grid-based accepted-set measure within $A_0 = [-1, 3]$. The measure increases strictly at every step as κ falls in all 30 estimator–sample-size–quantile sequences. At $n = 1000$ and $\tau = 0.50$, Oracle, Full, and DML medians rise from 0.200, 0.400, and 0.300 at $\kappa = 1$ to 3.900, 3.800, and 3.900 at $\kappa = 0.10$. Thus lower designed relevance is accompanied by a progressively larger median grid-acceptance measure, even when direct-at-truth acceptance moves closer to 0.95.

Quantile and estimator differences are visible before the weakest setting. At $n = 1000$ and $\kappa = 0.50$, DML's median measure ranges from 0.600 at $\tau = 0.50$ to 2.225 at $\tau = 0.10$, while Full ranges from 0.800 to 2.450 across the five quantiles. Increasing the sample size narrows the median accepted measure in 52 of 60 matched cells and leaves it unchanged in eight; it never widens it. For example, DML at $\tau = 0.50$ and $\kappa = 0.25$ falls from 3.050 to 2.025, whereas at $\kappa = 0.10$ it falls only from 4.000 to 3.900. The larger sample has a smaller or unchanged observed median accepted measure in these comparisons, with little localization under the weakest relevance setting.

The DML sequence at $n = 1000$ and $\tau = 0.50$ connects the four inferential summaries. At $\kappa = 1$, coverage is 0.776, the median accepted measure is 0.300, the full-grid acceptance rate is 0.000, and power at $\Delta = 0.50$ is 1.000. At $\kappa = 0.25$, the corresponding values are 0.948, 2.025, 0.040, and 0.286. At $\kappa = 0.10$, coverage remains near nominal at 0.942, but the median measure is 3.900, full-grid acceptance is 0.428, and power at $\Delta = 0.50$ is 0.094. Truth retention, the amount of the grid retained, acceptance of the entire grid, and rejection of a particular false value therefore answer different questions.

The measure cannot exceed four because it integrates grid acceptance only over $A_0 = [-1, 3]$. Similar values near four do not imply equivalent unrestricted confidence regions: the calculation does not examine candidates outside A_0, and distinct acceptance patterns can produce similar totals. Supplementary Figure C.1 records the frequency with which all 41 evaluated points are accepted. That event does not establish acceptance at every intermediate real value or anywhere outside A_0. Coverage is also a separate direct evaluation at truth. For DML, fresh pivotal draws mean that direct-truth coverage and stored-profile acceptance need not coincide even when truth is numerically a grid point.

Appendix A examines wider domains in a separate diagnostic with $n = 1000$, 100 replications, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and step 0.1 on $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. At $\tau = 0.50$ and $\kappa = 1$, median DML measure remains 0.300 on all three domains and A_2 boundary acceptance is zero. At $\kappa = 0.25$, it grows from 1.950 to 2.500 and 3.025, with A_2 boundary acceptance 0.500. At $\kappa = 0.10$, it grows from 3.900 to 7.500 and 10.325, and the A_2 boundary is accepted in 0.960 of usable profiles. Oracle and Full display the same broad weak-relevance pattern, with A_2 median measures 11.300 and 11.450 and boundary rates 0.830 and 0.850 in the corresponding cell.

These domain-expansion values come from a diagnostic run with 100 replications, not the main 500-replication experiment. They show near-stability within the investigated domains under strong relevance and continued failure to localise at intermediate and weak relevance. The diagnostic ends at A_2 and does not search the real line. Failure to localise over these domains is evidence of limited numerical informativeness, not proof of global unboundedness or non-identification.

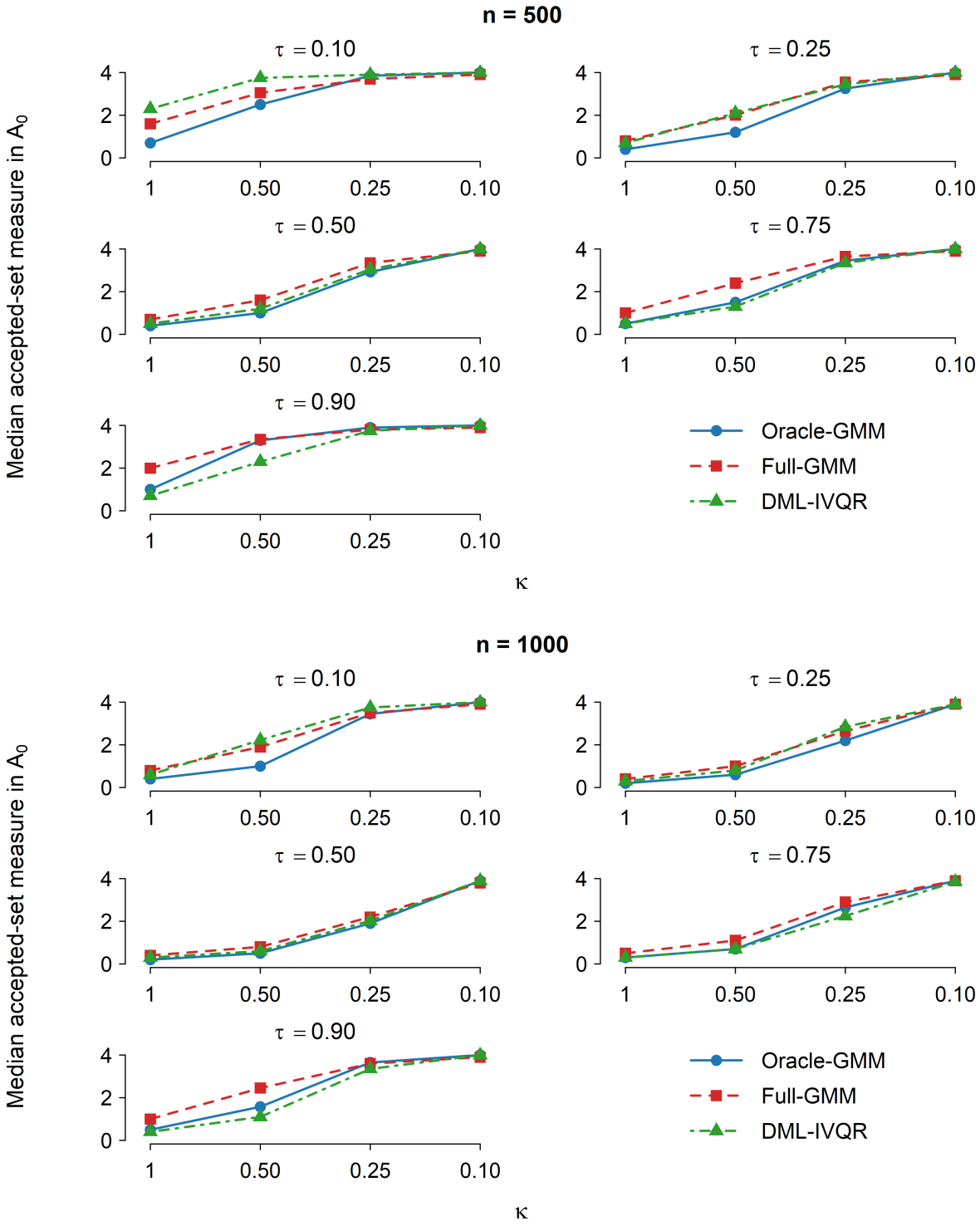


Figure 3. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

Note: Panels show $n = 500$ and $n = 1000$, using 500 replications per cell. Curves identify Oracle-GMM, Full-GMM, and DML-IVQR; quantile panels show τ. Plotted values are replication medians of the trapezoidal binary-grid-acceptance measure at each κ. The maximum is four, the width of A_0; this is not unrestricted confidence-region length. Lines join evaluated relevance settings as visual guides.

5 Discussion

The experiment distinguishes retaining a true candidate from excluding false effects. As relevance weakens, direct-at-truth acceptance can approach the nominal level while false-candidate rejection falls and most of the grid remains accepted. The strong-relevance undercoverage and documented power reversals in Section 4 remain material: neither nominal-looking acceptance nor a favourable point-estimation ranking establishes reliable and informative inference.

Oracle-GMM also loses accuracy and numerical localization despite knowing the designated relevant controls. High-dimensional nuisance estimation is therefore not the sole explanation for deterioration. The implementation-level RMSE comparisons have exceptions. Their comparison does not identify a regularization mechanism or establish uniform superiority. The larger sample improves point accuracy without improving every inferential measure; quantile patterns in bias and point error do not imply a corresponding calibration ordering. The complete cell evidence is mapped in Appendix D.2.

The ideal theory supplies motivation rather than a validated description of the executed tests. Orthogonality concerns nuisance sensitivity under its conditions (Chernozhukov, Chetverikov, et al. 2018); weak-identification-robust candidate testing concerns validity under its own conditions, not guaranteed power (Chernozhukov and Hansen 2008). Here the mixed residual density is non-finite (Appendix D.6), while the implementations use density proxies, differing residualization, same-sample nuisance fits and fresh pivotal penalties. Their nominal chi-square validity remains unestablished. Absence of cross-fitting alone does not prove invalidity, and these features were not varied separately to identify a cause of the observed distortions. Selected-case audits also document residual-rank and decision sensitivity; they neither provide a full-experiment error frequency nor replace historical results (Appendix D).

The practical implication is joint reporting of point accuracy, direct-test calibration diagnostics, false-candidate rejection and finite-grid informativeness. The trapezoidal accepted measure summarizes binary grid decisions; it does not locate continuous boundaries. Wider-domain diagnostics reveal limited localization over the examined ranges, not global unboundedness or non-identification. One DGP family, five quantiles, two sample sizes and three preserved implementations delimit the evidence. No particular numerical or statistical correction is shown to restore calibration, and two sample sizes establish no asymptotic rate. In applications, domain sensitivity and the maintained inference assumptions should accompany the nominal confidence level.

6 Conclusion

This controlled extension of the fixed-relevance design of Chen et al. (2021) reduces excluded-instrument signal while independent compensating noise preserves the joint treatment/outcome-disturbance distribution under the stated Gaussian construction. Comparing three preserved high-dimensional IVQR implementations shows why point accuracy, direct-test acceptance, false-candidate rejection and finite-grid informativeness must be assessed together.

The central empirical disagreement is between nominal-looking truth acceptance and useful exclusion of false effects. Retaining most of a grid while rejecting few false candidates offers little discrimination. Direct-at-truth acceptance is not membership coverage of the stored randomized grid set, and strong-relevance undercoverage remains a counterpoint to near-nominal rates elsewhere. Implementation-level RMSE advantages have documented exceptions and do not isolate regularization.

These findings establish neither implemented orthogonality nor validity of the nominal chi-square cutoff or automatic transfer of theoretical weak-identification robustness. The implementation and selected-case numerical limitations summarized in the Discussion restrict that link. Their certificates do not establish invariance of historical decisions or a full-experiment error frequency. Historical results are retained, finite grids establish no unrestricted confidence-set geometry, and no particular correction is shown to restore calibration.

Declarations

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Competing interests. The authors declare no financial or non-financial conflicts of interest.

Author contributions. Both authors contributed equally.

AI assistance. AI tools, including ChatGPT and OpenAI Codex, assisted with grammar checking, translation, manuscript editing and documentation. The authors state that AI was not used to develop or modify research code, simulations or statistical analysis.

Ethics and informed consent. The study uses simulated data without human participants or personal data. Ethics approval and informed consent were not required.

Data and code availability

The public repository is `https://github.com/U-Olim/DML-IVQR_Master_Thesis`. Repository visibility, commit `15b8b4dc95e3658f3b858a6f792ea2983299aaa6`, and directory listings for the main frozen CSVs, reporting summaries and package archives were checked through the GitHub API on 10 November 2026. That commit supplies the checked evidence archive, rather than identifying the present manuscript version. Readers can select it and follow the exact paths in Appendix D.2 to inspect both sample sizes, all quantiles and relevance settings without running an estimator. These are simulated data and archived computational outputs, not an external observational dataset. The freeze record names `ac9a9ccad027cde05488abb633e66cc02f87cc46`; that identifier was not resolvable in the present local object store and returned HTTP 404 from the repository API. It is retained as recorded provenance, not offered as a verified release checkout.

Historical rerunning is a different task. Source and configuration are under `thesis_extension/src/` and `thesis_extension/config/`; environment records, package archives and the captured Windows library are under `thesis_extension/environment/`. The recorded environment uses R 3.4.3 and pinned historical packages (Appendix D); their presence does not establish installation portability or exact random-number replay. Main-run worker-specific penalty states are not recoverable from the parent seed alone. Large grid-diagnostic profiles and saved primitives are local, Git-ignored artifacts identified in `thesis_extension/diagnostics/grid_sensitivity/GRID_RESULTS_MANIFEST.csv`; tracked diagnostic summaries remain available. Stage 2H pilot outputs under `journal_validation/stage2h/pilot/runs/` and its `journal_validation/stage2h/pilot/SHA256_MANIFEST.json` are also Git-ignored and are not part of the public tracked archive. No public deposit of these local artifacts or portable historical runtime is established here. This revision does not publish new results or replace historical fits with later certified numerical solutions.

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Supplementary appendices

Appendices A–D provide numerical-domain diagnostics, the controlled-design derivation, the coverage audit, and implementation notes. They remain inline; additional domain-diagnostic tables are supplied separately in the accompanying Technical Supplement.

A Numerical-Domain Diagnostics

The primary computational domain is $A_0 = [-1, 3]$, evaluated on 41 equally spaced candidate values with step size $h = 0.1$. Using the same domain for all sample sizes, quantiles, instrument-relevance settings, and estimators makes the accepted-set measures directly comparable across the simulation design.

The interval also contains the true structural quantile effects for all five investigated quantiles. Under the data-generating process,

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau),$$

so the true effects are approximately -0.282, 0.326, 1.000, 1.674, and 2.282 for $\tau = 0.10, 0.25, 0.50, 0.75$, and 0.90, respectively. All five structural truths lie strictly inside A_0; using their unrounded values, the minimum distance to either endpoint is approximately 0.718. The interval should therefore be understood as a common numerical search domain, not as an assumed theoretical parameter space.

A finite search domain can nevertheless become restrictive when instruments are weak and many candidate values are accepted. For this reason, a separate sensitivity diagnostic expands the search range from $A_0 = [-1, 3]$ to $A_1 = [-3, 5]$ and then to $A_2 = [-5, 7]$. Under strong instrument relevance, expanding the domain adds little accepted measure, indicating that accepted grid values are well localized within the investigated ranges, without establishing continuous accepted components. At intermediate relevance, the wider domains contain additional accepted values in some profiles. Under the weakest relevance setting, the accepted measure continues to increase as the domain is expanded and does not stabilize within A_2.

These results clarify the interpretation of the main accepted-set measure. A value close to four in A_0 means that the trapezoidal grid-acceptance summary is close to its maximum on the four-unit primary domain; it does not establish acceptance between evaluated points or unrestricted confidence-region length. Similarly, continued acceptance in the wider domains indicates poor numerical localization over the ranges examined, but it does not by itself establish that the theoretical confidence region is globally unbounded or that identification completely fails.

Table A.1 retains the key frozen domain-measure comparisons; Figure A.1 summarizes the expansion pattern.

The main diagnostic contains one singular-design error: replication 35, $\tau = 0.90$, $\kappa = 0.10$, DML-IVQR, candidate $a = -2.1$ in Stage 1. Stage 2 adds no new evaluation errors. Incomplete profiles receive missing aggregate measures; the affected expanded-domain cell therefore uses 99 usable replications, while other displayed cells use 100. The accompanying Technical Supplement retains Tables A.2–A.5: incremental measures, boundary frequencies, profile counts, and localization diagnostics, with their original numbering and entries. In these tables, Med. and Mean denote replication medians and means; R is the intended replication count, Succ. counts complete profiles, Fail counts incomplete profiles, and Stage 1/2 counts newly recorded evaluation errors. Boundary frequencies refer to acceptance of either endpoint; full-grid frequencies require acceptance at every evaluated point. A nonaccepted boundary does not rule out disconnected accepted components farther outside the examined interval.

Table A.1: Grid-domain diagnostic: accepted-set measures

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	1.00	Oracle-GMM	0.400	0.400	0.400	0.386	0.386	0.386
1000	0.10	1.00	Full-GMM	0.800	0.800	0.800	0.757	0.757	0.757

Continued on next page

Table A.1: Grid-domain diagnostic: accepted-set measures (Continued)

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	1.00	DML-IVQR	0.600	0.600	0.600	0.705	0.705	0.705
1000	0.10	0.25	Oracle-GMM	3.400	5.325	7.250	3.019	4.994	6.838
1000	0.10	0.25	Full-GMM	3.375	5.950	8.200	3.304	5.774	8.037
1000	0.10	0.25	DML-IVQR	3.750	5.950	7.950	3.607	6.040	8.245
1000	0.10	0.10	Oracle-GMM	4.000	7.900	11.900	3.745	7.381	11.002
1000	0.10	0.10	Full-GMM	3.900	7.800	11.600	3.785	7.394	10.975
1000	0.10	0.10	DML-IVQR	4.000	7.575	10.775	3.845	7.219	10.339
1000	0.25	1.00	Oracle-GMM	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.25	1.00	Full-GMM	0.400	0.400	0.400	0.429	0.429	0.429
1000	0.25	1.00	DML-IVQR	0.300	0.300	0.300	0.346	0.346	0.346
1000	0.25	0.25	Oracle-GMM	2.175	2.800	2.850	2.224	3.158	3.964
1000	0.25	0.25	Full-GMM	2.450	3.500	4.075	2.517	3.768	4.745
1000	0.25	0.25	DML-IVQR	2.750	4.650	6.225	2.630	4.130	5.712
1000	0.25	0.10	Oracle-GMM	3.900	7.700	11.500	3.492	6.699	9.872
1000	0.25	0.10	Full-GMM	3.825	7.600	11.100	3.595	6.891	10.083
1000	0.25	0.10	DML-IVQR	4.000	7.700	11.125	3.651	6.926	9.984
1000	0.50	1.00	Oracle-GMM	0.200	0.200	0.200	0.235	0.235	0.235
1000	0.50	1.00	Full-GMM	0.400	0.400	0.400	0.373	0.373	0.373
1000	0.50	1.00	DML-IVQR	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.50	0.25	Oracle-GMM	1.875	2.200	2.200	1.960	2.682	3.229
1000	0.50	0.25	Full-GMM	2.250	2.400	2.550	2.368	3.171	3.802
1000	0.50	0.25	DML-IVQR	1.950	2.500	3.025	2.124	3.020	3.942
1000	0.50	0.10	Oracle-GMM	3.825	7.600	11.300	3.521	6.640	9.687
1000	0.50	0.10	Full-GMM	3.900	7.625	11.450	3.536	6.718	9.817
1000	0.50	0.10	DML-IVQR	3.900	7.500	10.325	3.538	6.671	9.585
1000	0.75	1.00	Oracle-GMM	0.300	0.300	0.300	0.290	0.290	0.290
1000	0.75	1.00	Full-GMM	0.500	0.500	0.500	0.512	0.512	0.512
1000	0.75	1.00	DML-IVQR	0.300	0.300	0.300	0.279	0.279	0.279
1000	0.75	0.25	Oracle-GMM	2.825	4.200	4.850	2.628	3.945	4.990
1000	0.75	0.25	Full-GMM	2.950	4.450	5.150	2.827	4.267	5.478
1000	0.75	0.25	DML-IVQR	2.325	3.300	3.900	2.402	3.595	4.788
1000	0.75	0.10	Oracle-GMM	3.975	7.750	11.700	3.670	6.999	10.357
1000	0.75	0.10	Full-GMM	3.900	7.750	11.500	3.661	7.023	10.267
1000	0.75	0.10	DML-IVQR	3.900	7.350	10.375	3.554	6.691	9.736
1000	0.90	1.00	Oracle-GMM	0.500	0.500	0.500	0.519	0.519	0.519
1000	0.90	1.00	Full-GMM	1.000	1.000	1.000	1.026	1.031	1.031
1000	0.90	1.00	DML-IVQR	0.400	0.400	0.400	0.454	0.454	0.454
1000	0.90	0.25	Oracle-GMM	3.650	5.850	7.850	3.381	5.625	7.675
1000	0.90	0.25	Full-GMM	3.675	6.150	8.575	3.555	6.123	8.432
1000	0.90	0.25	DML-IVQR	3.450	5.400	7.300	3.114	5.249	7.374
1000	0.90	0.10	Oracle-GMM	4.000	7.900	11.800	3.712	7.241	10.771
1000	0.90	0.10	Full-GMM	3.900	7.700	11.500	3.817	7.448	11.031
1000	0.90	0.10	DML-IVQR	4.000	7.700	11.600	3.644	7.042	10.395

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

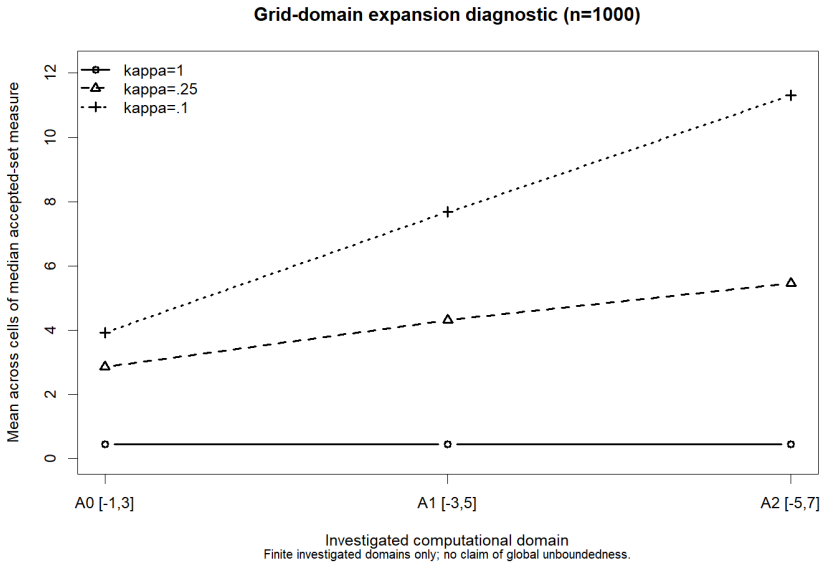


Figure A.1. Sensitivity of the accepted set to expansion of the computational domain

Note: Each plotted value is the mean across 15 estimator–quantile cells of the cell-specific median accepted-set measure for the stated κ and domain. The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

Figure C.1 in Appendix C complements these expansion diagnostics by reporting how often all candidates within the primary grid are accepted.

B Derivation and Properties of the κ Extension

The extension changes assignment while retaining the baseline primitives in Section 3.1. The excluded sum has variance two, ϵ_i has variance one, and $\mathrm{Cov}(u_i, \epsilon_i) = 0.3$. Introduce independent $w_i \sim \mathcal{N}(0,1)$ and write

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + c(\kappa)w_i + \epsilon_i, \qquad \kappa \in [0,1]. \tag{B.1}$$

Independence eliminates all cross-covariances between index components. Preserving the baseline variance requires

$$2\kappa^2 + c(\kappa)^2 + 1 = 3, \qquad c(\kappa) = \sqrt{2(1-\kappa^2)}, \quad 0 \le \kappa \le 1.$$

The nonnegative root is conventional; normal symmetry makes its sign distributionally immaterial. This coefficient is specific to the independent-noise construction. At $\kappa = 0.50$, the signal variance is 0.5, replacement variance is 1.5, and $c(0.50) = \sqrt{1.5} \approx 1.2247$. Calculations use the unrounded coefficient.

$$\begin{aligned} d_i^*(\kappa) &= \kappa(z_{1i} + z_{2i}) + \sqrt{2(1-\kappa^2)}\, w_i + \epsilon_i, \\ D_i(\kappa) &= \Phi\big(d_i^*(\kappa)\big)\,. \end{aligned} \tag{B.2}$$

Normality gives $d_i^*(\kappa) \sim \mathcal{N}(0,3)$, where 3 denotes variance. Thus for $0 < d < 1$,

$$\begin{aligned} \Pr[D_i(\kappa) \le d] &= \Pr\big[\Phi(d_i^*(\kappa)) \le d\big] \\ &= \Pr\big[d_i^*(\kappa) \le \Phi^{-1}(d)\big] \\ &= \Phi\left(\frac{\Phi^{-1}(d)}{\sqrt{3}}\right). \end{aligned} \tag{B.3}$$

The CDF is zero below zero and one above one. It is independent of κ, but is not Uniform$(0,1)$. Equal variance establishes this invariance because the index is normal, not for arbitrary distributions.

Write $Z_{ji} = z_{ji} + v_{ji} + C_{ji}$, where $C_{1i} = X_{2i} + X_{3i} + X_{4i}$ and $C_{2i} = X_{7i} + X_{8i} + X_{9i} + X_{10i}$. Independence yields

$$\mathrm{Cov}(Z_{ji}, d_i^*(\kappa)) = \kappa\,\mathrm{Cov}(z_{ji}, z_{1i} + z_{2i}) = \kappa, \qquad j = 1, 2.$$

The nonzero means of the control sums do not alter this covariance calculation. It concerns the latent index, and does not assert $\mathrm{Cov}(Z_{ji}, D_i(\kappa)) = \kappa$. Designed relevance is therefore distinct from a formal IVQR identification statistic.

Similarly, $\mathrm{Cov}(u_i, d_i^*(\kappa)) = \mathrm{Cov}(u_i, \epsilon_i) = 0.3$. Joint normality gives

$$\begin{pmatrix} u_i \\ d_i^*(\kappa) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 3 \end{pmatrix}\right). \tag{B.4}$$

The invariant mean vector and covariance matrix establish invariance of the entire joint distribution, hence of $(u_i, \Phi(d_i^*(\kappa)))$. Covariance alone would not suffice without joint normality. Reusing primitive draws aids matched comparisons but is not needed for this distributional result; individual treatment realizations can differ across relevance settings.

At fixed $d > 0$, $Y_i(d) = 1 + 5\sum_{j=1}^{7} X_{ji} + d(1 + u_i)$. Independence of u_i and X_i gives $Q_{Y_i(d)|X_i}(\tau) = 1 + 5\sum_{j=1}^{7} X_{ji} + d[1 + \Phi^{-1}(\tau)]$. The unchanged potential-outcome equation therefore preserves $\alpha_0(\tau)$. This is a structural quantile at fixed treatment, not an observational conditional quantile at realized endogenous treatment.

At $\kappa = 1$ the replacement noise vanishes and the baseline is recovered. At the conceptual endpoint $\kappa = 0$, $d_i^*(0) = \sqrt{2}w_i + \epsilon_i$, so the excluded primitive signal disappears. The weakest simulated setting retains signal variance $2(0.10)^2 = 0.02$. These endpoint statements establish properties of this construction, not global identification results for arbitrary IVQR models.

Table B.1 records the analytical variance allocation for all simulated settings.

Table B.1. Variance allocation in the κ-extension.

κ	Coefficient $\sqrt{2(1-\kappa^2)}$	Signal variance $2\kappa^2$	Noise variance $2(1-\kappa^2)$	Total index variance
1.00	0.0000	2.0000	0.0000	3.0000
0.50	1.2247	0.5000	1.5000	3.0000
0.25	1.3693	0.1250	1.8750	3.0000
0.10	1.4071	0.0200	1.9800	3.0000

Note: Values are calculated directly from the construction, not simulated. The disturbance ϵ_i contributes the remaining variance of one in every row.

C Coverage Audit and Inferential Informativeness

C.1 Direct-at-Truth Coverage Audit

The reported coverage diagnostic is direct candidate-test acceptance at the exact true structural effect $\alpha_0(\tau)$, independently of whether that value lies on the finite candidate grid. It is not literal membership coverage of the stored grid accepted set defined in Section 2.2; fresh pivotal draws can also distinguish the DML direct and stored-grid evaluations at the same candidate. The forensic audit checked 60,000 direct truth evaluations: 500 replications for each of 120 sample-size, quantile, instrument-strength, and estimator cells.

Wilson 95% Monte Carlo intervals exclude nominal 0.95 coverage in 54 of the 120 cells. The audit finds no missing or nonfinite true-parameter W statistics. The large coverage distortions reported in the main text are therefore neither plotting-grid artifacts nor consequences of missing-value handling. Wilson intervals are pointwise Monte Carlo uncertainty intervals, without simultaneous or multiplicity-adjusted significance claims. An individual rate's Monte Carlo standard error is not a standard error for a difference between correlated simulation results.

These checks do not establish correct theoretical specification, validate the asymptotic reference distribution for this DGP, or establish nominal finite-sample calibration.

C.2 Coverage-Audit Summary

Table C.1 summarizes the audit by estimator and sample size. Each row contains 20 quantile–κ cells. Recalculation from integer acceptance counts reproduces 53 cells with an upper Wilson endpoint below 0.95 and one with a lower endpoint above 0.95, giving the reported total of 54. Comparisons use unrounded endpoints and the same 500-evaluation denominator in every cell.

Table C.1. Direct-at-truth coverage audit.

Estimator	n	Cells	Coverage min.	Coverage max.	Upper < 0.95	Lower > 0.95	Missing/ nonfinite
Oracle-GMM	500	20	0.858	0.954	6	0	0
Oracle-GMM	1000	20	0.860	0.964	8	0	0
Full-GMM	500	20	0.442	0.962	9	0	0
Full-GMM	1000	20	0.502	0.962	12	0	0
DML-IVQR	500	20	0.658	0.976	8	1	0
DML-IVQR	1000	20	0.498	0.968	10	0	0

Note: Each row covers 20 quantile–κ cells, with 500 direct-at-truth evaluations per cell. Coverage is reported as a proportion. The upper/lower columns count cells whose unrounded Wilson upper endpoint is below 0.95 or lower endpoint is above 0.95. Intervals use integer acceptance counts, the verified denominator, and the two-sided 95% Wilson formula without continuity correction. They are pointwise Monte Carlo intervals, not simultaneous or multiple-testing-adjusted intervals; exclusion counts are descriptive audit summaries. Missing/nonfinite counts concern direct-at-truth W evaluations (missing values are counted once).

C.3 Full-Domain Grid Acceptance

The domain-expansion diagnostics in Appendix A are complemented by Figure C.1, which reports how often every evaluated point in the primary domain is accepted. This frequency complements direct-at-truth acceptance by measuring retention of competing primary-grid candidates.

The heatmap uses stored grid-profile evaluations, whereas coverage uses a separate evaluation directly at truth. These are distinct computations; in DML-IVQR, fresh pivotal draws can differ between them, so their acceptance events need not coincide even when truth is a grid point. Full-domain grid acceptance refers only to the 41 evaluated points in $A_0 = [-1, 3]$, not to acceptance of the real line or global unboundedness of an unrestricted confidence region.

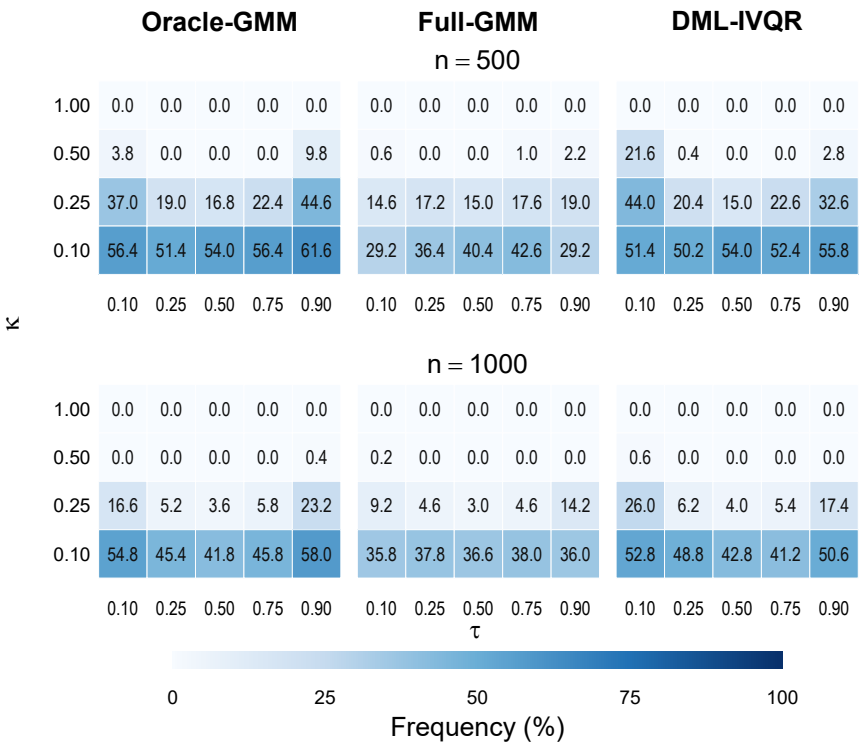


Figure C.1. Full-domain grid acceptance frequencies.

Note: Each cell reports the percentage of 500 complete replications accepting all 41 evaluated grid points in $A_0 = [-1, 3]$, with step size $h = 0.1$. Panel columns identify estimators; panel rows show $n = 500$ and $n = 1000$. Within each panel, columns show quantiles τ and rows show relevance κ, declining from 1.00 to 0.10. Darker shading on the shared 0–100% scale means more frequent full-domain grid acceptance. This event neither denotes acceptance of the entire real line nor establishes an unbounded unrestricted confidence region.

D Reproducibility and Implementation Notes

D.1 Environment, records, and reproducibility limits

The computational archive and checked evidence revision are identified in the data and code availability statement. They do not establish bit-for-bit regeneration from a seed on arbitrary systems. Appendix D.2 links the scientific findings to their frozen evidence; the full repository inventory retains source paths and the unresolved historical freeze record.

The canonical environment is R 3.4.3 (30 November 2017) on 64-bit Windows, using five workers. The captured attached packages are `doSNOW` 1.0.16, `snow` 0.4-2, `iterators` 1.0.8, `foreach` 1.4.3, `mvtnorm` 1.0-6, `hqreg` 1.4, `hdm` 0.2.0, `quantreg` 5.34, and `SparseM` 1.77. Environment session and library manifests are under `thesis_extension/environment/`. The main simulation uses seed 675; the grid diagnostic also uses seed 675, with saved primitives reloaded in Stage 2; calibration uses seed 20260813. Worker-specific RNG streams are not explicitly initialized in the preserved parallel DML procedure, so the parent seed alone does not establish exact regeneration of every penalty draw.

Frozen CSVs permit deterministic reporting without re-estimation. The full evidence inventory records historical manifests and figure-check scopes, which do not certify a later source revision. The coverage summary uses integer direct-at-truth counts; the variance illustration is analytical. No historical validation or simulation is rerun for this document.

The original Monte Carlo results remain outputs of the preserved historical executable implementation. Later selected-case Stage 2G audits identified sensitivity of residual ranks and test decisions to computational details: historical Policy A and the certified-outcome Policy C path differed in 5 of the 24 saved candidate problems examined. Policy C uses exact outcome residual ranks with the retained density/projection pathway; these panel results do not replace the historical Monte Carlo results. Numerical certificates establish properties of specified fixed objectives and input values, with decision bounds conditional on the stated exported quantities; they do not establish invariance of every historical Monte Carlo decision or certify all adaptive calibration trajectories. The selected panels do not establish an error frequency for the full Monte Carlo experiment.[1]

D.2 Frozen-evidence map

Keys E1–E6 below connect the findings to frozen evidence. The full locator inventory accompanies this proposal as `EVIDENCE_INVENTORY.md` under `journal_validation/publication_review/lssp_optimization/`. It retains all original paths, revision records and access limitations; it does not recompute results or establish public access to this proposed document. The checked evidence revision is distinct from the present manuscript version and the unresolved historical freeze identifier.

Finding	Manuscript location	Evidence key
Designed relevance ordering	Section 4.1	E1
Point-error deterioration and qualified estimator rankings	Section 4.2	E2
Truth acceptance and false-candidate rejection	Sections 4.3–4.4	E3
Nominal-looking acceptance with little grid exclusion	Abstract; Section 4	E2–E4
Domain sensitivity and finite numerical scope	Appendix A	E5
Selected-case rank and decision sensitivity	Appendix D	E6

E1 identifies the 100-replication relevance calibration; E2 the main performance summaries, replication records and profiles for both sample sizes; E3 the direct truth tests and false-candidate rejection records; E4 grid informativeness; E5 the separate 100-replication domain expansion, including its failed profile;

[1] See `journal_validation/stage2g/full_path_validation/FULL_PATH_REPORT.md`, "Complete 24-problem decision comparison" and "Required conclusions"; and `journal_validation/stage2g/deterministic_implementation/DETERMINISTIC_IMPLEMENTATION_REPORT.md`, "Mathematical equivalence: established scope".

and E6 the selected-case numerical-sensitivity reports. The inventory supplies the exact CSV, table, figure and report locators for each key. Main frozen summaries are available at the checked revision described in the availability statement; large diagnostic profiles/primitives and Stage 2H raw outputs remain local-only. A locator is not a guarantee of public access or full stochastic replay.

D.3 Historical evaluation sequence

The operational source is `thesis_extension/src/run_extension.R`, with candidate routines in `thesis_extension/src/author_bc_dml_wrapper.R` and `thesis_extension/src/author_oracle_wrapper.R`. One replication proceeds as follows.

1. Generate one set of primitives; construct a sample at each relevance setting. Within that setting all three estimators share (Y, D, X, Z). Reusing primitives across settings does not hold realized (Y, D) fixed.

2. For each quantile and estimator, profile the ordered 41-point primary grid. At each DML candidate a, call `lambda_BC_author` afresh (1000 artificial uniform columns), then fit `quantreg::rq(y-a*D~X, method="lasso")` with the resulting penalty and an intercept. Refit nuisances at each candidate; there is no post-Lasso refit. Oracle and Full instead call unpenalized `quantreg::rq` with their designated controls.

3. Form candidate outcome residuals and the proxy `dnorm(e, mean(e), var(e))`. Residualize instruments using the DML transformed regressions or benchmark matrix rule below; numerical termination and last-used penalty conventions are specified once in Appendix D.5.

4. Apply the literal computed-residual rule $r_i = \mathbf{1}\{\widehat{e}_i \leq 0\}$, with no residual tolerance. Set $s_i = \tau - r_i$, $g = \sum_i \widehat{\Psi}_i s_i$ and $S = \sum_i \widehat{\Psi}_i \widehat{\Psi}_i' s_i^2$. The statistic is $g' S^{-1} g$, equivalent to the normalized definition in Section 2. Use direct `solve` and the retained nominal chi-square cutoff; singular or non-finite evaluations are failures, not acceptances.

5. Retain the first grid minimum for the point estimate and the stored grid decisions for informativeness summaries. After profiling, separately evaluate the true effect and six specified false candidates. DML again invokes its candidate routine, with fresh pivotal draws, in a separate five-worker evaluation. Direct-at-truth acceptance and false-candidate rejection therefore do not reuse stored grid statistics, even when candidates coincide. Failed profiles receive no point estimate; the documented failure conventions remain in force.

The archived estimator includes both sample randomness and algorithmic penalty randomness. Common observed data do not imply common pivotal draws across estimators, candidates, or repeated calls. Worker-specific RNG stream assignments and the full main-run candidate penalty states were not archived as a replay specification; seed 675 alone neither reproduces every historical penalty nor establishes independence of worker streams. Saved Stage 2G cases with exported states have a narrower replay scope. The conventional Monte Carlo standard-error summaries do not verify historical stream independence. Frozen CSV inspection and selection of existing table/figure assets require no stochastic rerun; they do not recover missing RNG states.

D.4 Penalty, density proxy, and residualization

For each candidate, b_0 is the fitted outcome intercept and b the slopes. In the preserved `rq.fit.lasso` augmented-observation implementation, the unscaled check-loss sum receives a penalty with factor $1/2$. The vector $\lambda^{rq}(a)$ is the supplied routine argument, including an intercept penalty. There is no post-Lasso refit. Let $s_k = (n^{-1} \sum_i X_{ik}^2)^{1/2}$; U_{ir} are independent artificial Uniform$(0, 1)$ draws, with $r = 1, \ldots, 1000$, and $\widehat{Q}_{0.9}$ is their empirical pivotal-statistic 90th percentile. Fresh draws are made on every penalty call. This is algorithmic randomness, not sample splitting.

In the following formulas, $\phi(e;\mu,\sigma)$ denotes a normal density parameterized by standard deviation. The proxy supplies $\widehat{\sigma}^2_\epsilon$ in that argument: the preserved R call is `dnorm(e, mean(e), var(e))`. It evaluates a fitted-normal density at residuals rather than estimating the theoretical conditional density at zero. The sample-variance convention is

$$\widehat{\sigma}^2_\epsilon(a) = \frac{1}{n-1}\sum_{i=1}^{n}\left[\widehat{\epsilon}_i(a) - \widehat{\mu}_\epsilon(a)\right]^2.$$

This variance, not its square root, is supplied as the normal scale; the historical convention is described, not corrected. The instrument penalty is $\mathcal{P}_{j,a}(\gamma_j) = \sum_k \lambda^{\Gamma}_{jk}(a)|\gamma_{jk}|$, using default iterative heteroskedastic loadings. Its ordinary intercept is unpenalized; no post-Lasso refit is performed.

$$\begin{pmatrix}\widehat{b}_0(a)\\ \widehat{\beta}(a)\end{pmatrix} = \underset{b_0,b}{\arg\min}\left\{\sum_{i=1}^{n}\rho_\tau\left(Y_i - D_i'a - b_0 - X_i'b\right) + \frac{1}{2}\left[\lambda^{rq}_0(a)|b_0| + \sum_{k=1}^{p}\lambda^{rq}_k(a)|b_k|\right]\right\}. \tag{D.1}$$

$$\begin{aligned} T_r &= \max_{1\le k\le p}\left|\frac{\sum_{i=1}^{n} X_{ik}\left[\tau - \mathbf{1}\{U_{ir}<\tau\}\right]}{s_k\sqrt{\tau(1-\tau)}}\right|, \\ \lambda^{rq}(a) &= 2\,\widehat{Q}_{0.9}\left(\{T_r\}_{r=1}^{1000}\right)\sqrt{\tau(1-\tau)}(1, s_1, \ldots, s_p)'. \end{aligned} \tag{D.2}$$

$$\begin{aligned} \widehat{\epsilon}_i(a) &= Y_i - D_i'a - \widehat{b}_0(a) - X_i'\widehat{\beta}(a), \\ \widehat{\mu}_\epsilon(a) &= \frac{1}{n}\sum_{i=1}^{n}\widehat{\epsilon}_i(a), \\ \widehat{f}_i(a) &= \phi\left(\widehat{\epsilon}_i(a); \widehat{\mu}_\epsilon(a), \widehat{\sigma}^2_\epsilon(a)\right), \end{aligned} \tag{D.3}$$

$$\begin{pmatrix}\widehat{r}_{j0}(a)\\ \widehat{\Gamma}_{\mathrm{imp},j}(a)\end{pmatrix} = \underset{(r_{j0},\gamma_j)}{\arg\min}\left\{\sum_{i=1}^{n}\left[\sqrt{\widehat{f}_i(a)}Z_{ji} - r_{j0} - \sqrt{\widehat{f}_i(a)}X_i'\gamma_j\right]^2 + \mathcal{P}_{j,a}(\gamma_j)\right\}. \tag{D.4}$$

$$\begin{aligned} \widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{r}_{j0}(a) - X_i'\widehat{\Gamma}_{\mathrm{imp},j}(a), \\ \widehat{\Psi}_i(a) &= Z_i - \widehat{r}_0(a) - \widehat{\Gamma}_{\mathrm{imp}}(a)X_i. \end{aligned} \tag{D.5}$$

$$\begin{pmatrix}\widehat{b}_{0,\mathrm{orc}}(a)\\ \widehat{\beta}_{\mathrm{orc}}(a)\end{pmatrix} = \underset{b_0,b}{\arg\min}\sum_{i=1}^{n}\rho_\tau\left(Y_i - D_i'a - b_0 - X_{S_0,i}'b\right). \tag{D.6}$$

$$\begin{pmatrix}\widehat{b}_{0,\mathrm{full}}(a)\\ \widehat{\beta}_{\mathrm{full}}(a)\end{pmatrix} = \underset{b_0,b}{\arg\min}\sum_{i=1}^{n}\rho_\tau\left(Y_i - D_i'a - b_0 - X_i'b\right). \tag{D.7}$$

$$\begin{aligned} \widehat{M}_e(a) &= \frac{1}{n}\sum_{i=1}^{n} Z_i X_{e,i}'\widehat{f}_{e,i}(a), \qquad \widehat{J}_e(a) = \frac{1}{n}\sum_{i=1}^{n} X_{e,i}X_{e,i}'\widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a)\widehat{J}_e(a)^{-1}, \qquad \widehat{\Psi}_{e,i}(a) = Z_i - \widehat{\delta}_e(a)X_{e,i}. \end{aligned} \tag{D.8}$$

The first five equations describe DML-IVQR. The last three describe Oracle-GMM and Full-GMM, with $S_0 = \{1,\ldots,10\}$, $X_{\mathrm{orc},i} = X_{S_0,i}$, and $X_{\mathrm{full},i} = X_i$. Their quantile regressions include an intercept, but their matrix residualization includes no constant regressor. DML retains the ordinary intercept of a transformed regression and subtracts it from the original instrument without dividing by $\sqrt{\widehat{f}_i(a)}$. It is therefore not the exact sample counterpart of the ideal density-weighted intercept. All procedures use same-sample nuisance fits and scores; cross-fitting is absent. The comparison does not isolate regularization from these residualization differences.

D.5 Objectives and finite historical solver outputs

The argmin displays above specify optimization objectives; the archived coefficients are finite routine outputs, not certified exact minimizers. The DML outcome route is `quantreg::rq(..., method="lasso")` / `rq.fit.lasso` in `quantreg` 5.34. It uses augmented observations and the penalized intercept described above. Oracle and Full use the package's default unpenalized `rq` route. Computed residual signs, including exact floating-point zeros, feed the literal ≤ 0 rule; small objective errors alone do not establish rank or decision invariance.

Instrument fitting uses `hdm` 0.2.0 `rlasso` with `post=FALSE`, the default heteroskedastic, X-independent loading calculation and an ordinary unpenalized intercept. In the retained source, outer calibration has `numIter=15` and `tol=1e-5`; it stops when the absolute change in the estimated residual standard deviation is below the tolerance or the iteration budget is exhausted. A zero-selected-coefficient branch can return early. Inner `LassoShooting.fit` uses `maxIter=1000`, `optTol=1e-5` and `zeroThreshold=1e-6`; its loop starts at one and runs while the counter is below the maximum, stopping early when the sum of absolute coefficient changes falls below `optTol`. The returned small coefficients are thresholded. These are finite algorithmic conventions, not accuracy certificates. The pinned source is in `thesis_extension/environment/package_archives_R343/hdm_0.2.0.tar.gz` (`hdm/R/rlasso.R` and `hdm/R/LassoShooting.fit.R` inside that archive).

The outer routine updates loadings after a finite shooting fit. Its reported next penalty need not be the last penalty actually used to produce those coefficients. The later contract `journal_validation/stage2g/deterministic_implementation/IMPLEMENTATION_CONTRACT_v1.md` distinguishes preservation of the adaptive trajectory from refinement of the final centered objective at the last-used penalty. Refined final coefficients are not fed back into that trajectory under its Policy B contract; recalibration is a different pathway. Those selected-case numerical investigations do not establish exact attainment by historical fits. No certified solution, altered residual threshold or revised penalty path replaces any archived estimator result here. The selected-panel decision-sensitivity disclosure above remains applicable.

D.6 Retained density-divergence argument for the original DGP

This argument is reproduced from Section 3 of `journal_validation/stage2h/readiness/BOUNDED_DGP_DERIVATION.md`, specifically its analysis of the original $D = \Phi(T)$ construction; no bounded-treatment results are substituted. Put $\rho = 0.3$, $q_\tau = \Phi^{-1}(\tau)$, $Q_j = Z_j - h_j(X) = z_j + v_j$, and $\mu_Q = \kappa(Q_1 + Q_2)/2$. The observed instrument shifts $h_j(X)$ are given in the DGP. Since $Q \sim \mathcal{N}(0, 2I_2)$, Gaussian conditioning in the retained derivation gives

$$T \mid Q \sim \mathcal{N}(\mu_Q, v_\kappa), \qquad v_\kappa = 3-\kappa^2, \qquad T \mid u = q_\tau, Q \sim \mathcal{N}(\mu_Q+\rho q_\tau, s_\kappa^2), \quad s_\kappa^2 = v_\kappa-\rho^2 = 2.91-\kappa^2.$$

Controls are independent of (T, u, Q), and conditioning on (X, Z) identifies Q. For the structural residual $e_\tau = \Phi(T)(u - q_\tau)$, Bayes' formula and $u \perp (X, Z)$ yield the mixed-density identity

$$f_{e_\tau|X,Z}(0) = \phi(q_\tau)\,\mathrm{E}\left[\frac{1}{\Phi(T)}\,\middle|\, u = q_\tau, Q\right]. \tag{D.9}$$

Here $s_\kappa^2 > 1$ at every investigated relevance setting (1.91 at $\kappa = 1$ and 2.90 at $\kappa = 0.10$). As $t \to -\infty$, $1/\Phi(t) \sim |t|/\phi(t)$. Against the conditional normal density, the integrand is proportional in its tail to

$$|t| \exp\left\{\frac{1 - 1/s_\kappa^2}{2} t^2 + O(t)\right\},$$

whose negative-tail integral diverges. Thus the original $f_{e_\tau|X,Z}(0)$ is infinite. Mixing conditional only on X instead gives Gaussian variance $3 - \rho^2 = 2.91 > 1$ and the same divergence. These are densities that integrate over treatment, not $f_{e_\tau|D,X,Z}(0)$ at a realized positive treatment value. Nor are they the historical fitted-normal proxy. This retained regularity limitation does not establish which feature causes any particular Monte Carlo rejection pattern and does not validate or repair nominal inference.

D.7 Score algebra and its conditions

The ideal nuisance vector and derivative identities are retained below. They presume finite density-weighted moments and permissible differentiation, conditions not established for the simulation DGP's non-finite true residual density. They are not implementation-level assertions. The executed proxy, residualization and nuisance fits do not inherit these properties automatically, and the chi-square cutoff is used nominally without an established validity result for the preserved implementations under this DGP. Same-sample fitting requires its own nuisance and generated-score justification; absence of cross-fitting alone does not prove invalidity.

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \operatorname{vec}\{\delta(a)\} \end{pmatrix}, \qquad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \operatorname{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \tag{D.10}$$

$$\mathrm{E}\big[\{\tau - \mathbf{1}\big[Y_i \le D_i'a + c(a) + X_i'\beta(a)\big]\}\,\widetilde{X}_i\big] = 0. \tag{D.11}$$

$$\begin{aligned} M_0 &= \mathrm{E}\big[Z_i\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ J_0 &= \mathrm{E}\big[\widetilde{X}_i\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \tag{D.12}$$

$$\begin{aligned} \partial_{(c,\beta')}\,\mathrm{E}[g_\tau(W_i;\alpha_0(\tau),\eta)]\big|_{\eta=\eta_0} &= -\mathrm{E}\big[(Z_i - \delta_0\widetilde{X}_i)\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big] \\ &= -\left[M_0 - \delta_0 J_0\right] = 0. \end{aligned} \tag{D.13}$$

$$\partial_{\delta_j}\,\mathrm{E}\big[g_{\tau,j}(W_i;\alpha_0(\tau),\eta)\big]\big|_{\eta=\eta_0} = -\mathrm{E}\big[\{\tau - \mathbf{1}\big[Y_i \le D_i'\alpha_0(\tau) + c_0(\tau) + X_i'\beta_0(\tau)\big]\}\,\widetilde{X}_i'\big] = 0. \tag{D.14}$$

The first derivative cancels by $\delta_0 = M_0 J_0^{-1}$; the second cancels by the profiled quantile moment. Orthogonality concerns local nuisance sensitivity, and does not establish identification or finite-sample calibration.